QUIZ

Tap image for quiz

Figure 4

How do medulloblastomas appear on histology?

What Are Neuronal Tumors?

Neuronal tumors comprise abnormal glial and neuronal elements. They can be classified as pure neuronal cell tumors (eg, gangliocytomas, which are exceedingly rare and not discussed here) or mixed neuronal cell tumors (eg, ganglioglioma). Overall, neuronal tumors make up about 1% of all brain neoplasms.

Most neuronal tumors are located within the brain parenchyma, producing characteristic deficits based on location. As neuronal tumors grow, they disrupt the normal brain architecture and thus the neuronal networks. As a result, patients often present with seizures. Neuronal cell tumors are usually low grade and slow growing, so they have a favorable prognosis. They are usually cured with surgery alone.

As the name suggests, **gangliogliomas** are mixed neuronal cell tumors that contain a disorganized arrangement of ganglion cells and glia (indicating neuronal support cells). These tumors are often slow growing and therefore usually considered benign (World Health Organization [WHO] grade I).

Gangliogliomas have an incidence of 4% of all primary brain tumors in children and 1% of all primary brain tumors in adults. Up to 10% can become malignant. They tend to occur in the temporal lobes and present as new-onset seizures in children and young adults. Treatment of these cells. You order a CT scan, which reveals a large tumor in the left internal auditory canal affecting the vestibular portion of CN VIII. In consultation with a neurosurgeon, EB elects to have surgical resection. At his 3-month follow-up visit, his hearing loss persists, but his tinnitus is markedly improved. “I can live with the hearing loss, but that ringing in the ears was a killer. I’m so glad it’s gone,” EB says.

Summary

- Schwannomas are benign, encapsulated proliferations of neoplastic Schwann cells.
- Microscopically, schwannomas demonstrate dense and loose areas known as Antoni A and B, respectively.
- Verocay bodies are characteristic features of schwannomas, exhibiting palisading nuclear and anuclear regions.
- Sporadic and familial schwannomas arise because of loss of function of the merlin protein.
- Neurofibromas are benign proliferations of Schwann cells that also contain other cell types.
- The three distinct subtypes of neurofibromas are localized cutaneous, plexiform, and diffuse.
- Neurofibromas form because of loss of function of the neurofibromin protein.
- Neurofibromin acts as a negative regulator of the Ras cell growth pathway.
- Malignant peripheral nerve sheath tumors (MPNSTs) are rare but serious cancers, and patients with NF1 are particularly at risk; MPNSTs may arise from normal nerves or existing plexiform neurofibromas.
- NF1 presents with Lisch nodules, inguinal and axillary freckling, café au lait spots, optic gliomas, neurofibromas, and other abnormalities.
- NF2 presents with cranial nerve schwannomas, intracranial meningiomas, cataracts, and cutaneous tumors.
- NF1 results from mutation of the gene encoding neurofibromin.
- NF2 results from mutations of the gene encoding merlin.
- NF1 requires close follow-up to evaluate and manage the disease; patients have a decreased average lifespan.
- NF2 requires appropriate treatment of tumors and often leads to early death.

Review Questions

1. A 24-year-old male who initially presented with bilateral hearing loss and tinnitus demonstrates bilateral lesions within the internal acoustic canal on imaging. This patient most likely has a genetic defect affecting which chromosome?

A. Chromosome 7

B. Chromosome 11

C. Chromosome 17

D. Chromosome 21

E. Chromosome 22

Explanation (requires correct answer)